「Compact space.

1.a). Let $\mathcal{U}, \mathcal{U}'$ be Topologies on X, with $\mathcal{U} \subseteq \mathcal{U}'$.
If $(X, \mathcal{U}')$ is Compact, then $(X, \mathcal{U})$ is Compact.

Prof. Let $\mathcal{O} \subseteq \mathcal{U}$ be a open Cover of X. Then, since $\mathcal{O} \subset \mathcal{U} \subseteq \mathcal{U}'$ and $(X, \mathcal{U}')$ Compact
$\mathcal{O}$ has finite subcover. ☒

1.b). If X is Compact T_2 under both $\mathcal{U}$ and $\mathcal{U}'$,
then either $\mathcal{U}$ and $\mathcal{U}'$ are equal or they are Not comparable.

Prof. If $\mathcal{U} = \mathcal{U}'$, then there is nothing to prove. Suppose $\mathcal{U} \neq \mathcal{U}'$.
Enough to show that neither $\mathcal{U} \subset \mathcal{U}'$ nor $\mathcal{U}' \subset \mathcal{U}$.
Suppose that $\mathcal{U} \subset \mathcal{U}'$. Then, we can choose an open $\mathcal{U} \in \mathcal{U}' \setminus \mathcal{U}$.
Now, $X \setminus \mathcal{U}$ is closed in $\mathcal{U}'$ but not $\mathcal{U}$. Since $\mathcal{U}'$ is Compact T_2,
$X \setminus \mathcal{U}$ is Compact set of $\mathcal{U}'$. Since $\mathcal{U} \subset \mathcal{U}'$, for any open cover $\mathcal{O} \subseteq \mathcal{U}$ for $X \setminus \mathcal{U}$,
it has finite subcover. Therefore, $X \setminus \mathcal{U}$ is compact in $(X, \mathcal{U})$, thus Closed.
But, it is Contradiction. Similarly, $\mathcal{U}' \not\subset \mathcal{U}$.

2.a). Every Subspace in Co-finite Topology $(\mathbb{R}, \mathcal{U}_f)$ is Compact.

b). In Co-Countable $(\mathbb{R}, \mathcal{U}_c)$, is $[0,1]$ Compact?

Prof. Let $A \subseteq \mathbb{R}$ be a subset. Let $\mathcal{O} = \{\mathbb{R} \setminus F \mid F \text{ is finite}\}$ be an open Cover for A. That is,

$$A \subseteq \bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} \iff \bigcap_{\mathcal{U} \in \mathcal{O}} \mathbb{R} \setminus \mathcal{U} \subseteq A^c$$

But since $\mathbb{R} \setminus \mathcal{U}$ is finite for each $\mathcal{U} \in \mathcal{O}$, there are finite number of $\mathcal{U}_1, \cdots, \mathcal{U}_n$ s.t

$$\bigcap_{i=1}^{n} \mathbb{R} \setminus \mathcal{U}_i \subseteq A^c \iff A \subseteq \bigcup_{i=1}^{n} \mathcal{U}_i$$

Therefore A is Compact. (More Specifically, for any distinct finite set F_1 and F_2,
$F_1 \cap F_2$ has number of less than $\min\{\#(F_1), \#(F_2)\}$.)
But, in the Co-Countable, $[0,1]$ is Not compact, because
Construct sequences; denote $\mathbb{Q} \cap [0,1] = \{x_n \mid n \in \mathbb{N}\}$, and let $\mathcal{U}_n = (\mathbb{R} \setminus \mathbb{Q}) \cup \{x_n\}$
Then, $\bigcup_{n=1}^{\infty} \mathcal{U}_n \supseteq [0,1]$ but it has no finite subcover.

⌈3. Finite union of Compact Subspace is Compact.

Proof. Let $K_1, \cdots, K_n \subseteq X$ be Compact Subsets. Let $\mathcal{O} \subseteq \mathcal{P}(X)$ be an open cover of $\bigcup_{i=1}^{n} K_i$.
Since $\bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} \supseteq \bigcup_{i=1}^{n} K_i \supseteq K_j$ $(1 \le j \le n)$, for each j, $\mathcal{O}$ has a finite subcover for K_j,
denote $\mathcal{U}_j^1, \cdots, \mathcal{U}_j^{r_j} \in \mathcal{O}$. Now, clearly

$$\bigcup_{i=1}^{n} K_i \subseteq \bigcup_{j=1}^{n} \bigcup_{m=1}^{r_j} \mathcal{U}_j^m$$

⌈4. Every Compact Subspace in a Metric space is closed bounded.
Give the example that the Converse fails.

Proof. Closed is given by : Compact in T_2 is closed, because
Let X be a Hausdorff Space, and $K \subseteq X$ be a Compact Subset.
Let $x \in X \setminus K$ be fixed. Now, for each $y \in K$, there exists disjoint open $\mathcal{U}_y^x, \mathcal{U}_y$ s.t

$$x \in \mathcal{U}_y^x, \quad y \in \mathcal{U}_y, \quad \mathcal{U}_y^x \cap \mathcal{U}_y$$

Now, $\{\mathcal{U}_y \mid y \in K\}$ is open cover of K, therefore there is a finite subcover
$\{\mathcal{U}_{y_i} \mid i = 1, 2, \cdots, r_x\}$ and open neiborhoods of x $\{\mathcal{U}_{y_i}^x \mid i = 1, \cdots, r_x\}$.
Put $V_x = \bigcap_{i=1}^{r_x} \mathcal{U}_{y_i}^x$. Then, V_x is open neiborhood and $K \cap V_x = \phi$, thus $V_x \subseteq K^c$.
(More specifically, $K \subseteq \bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}$ and $\left(\bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}\right) \cap V_x = \bigcup_{i=1}^{r_x} (\mathcal{U}_{y_i} \cap V_x) = \bigcup_{i=1}^{r_x} \phi = \phi$.

And boundedness is clear.

Counterexample: Let $X = \{\frac{1}{n} \mid n \in \mathbb{N}\}$ discrete Topology.
Then, it is closed itself, and bounded, but not Compact.

「5. Let X be a Hausdorff, and $A, B \subseteq X$ be disjoint Compact subsets.
Then, there exist disjoint open sets $\mathcal{U}$ and V Containing A and B, respectively.

Proof. Let $x \in A$ be fixed. Then, for each $y \in B$, there exist opens $\mathcal{U}_{(x,y)}$ and $V_{(x,y)}$ s.t

$$x \in \mathcal{U}_{(x,y)},\quad y \in V_{(x,y)},\quad \mathcal{U}_{(x,y)} \cap V_{(x,y)} = \phi.$$

Since $\{V_{(x,y)} \mid y \in B\}$ is open cover of B, there exist a finite subcover $\{V_{(x,y_i)} \mid i = 1, 2, \cdots, r_x\}$. Now, $\mathcal{U}_x = \bigcap_{i=1}^{r_x} \mathcal{U}_{(x,y_i)}$ is open neiborhood of x, $V_x = \bigcup_{i=1}^{r_x} V_{(x,y_i)}$ is open set containg B, and $\mathcal{U}_x \cap V_x = \phi$. because

$$\mathcal{U}_x \cap V_x = \mathcal{U}_x \cap \left(\bigcup_{i=1}^{r_x} V_{(x,y_i)}\right) = \bigcup_{i=1}^{r_x} \left(\mathcal{U}_x \cap V_{(x,y_i)}\right) = \bigcup_{i=1}^{r_x} \phi = \phi$$

In Summary, for each $x \in A$, there are two disjoint opens $\mathcal{U}_x, V_x$ s.t

$$x \in \mathcal{U}_x,\quad B \subseteq V_x,\quad \mathcal{U}_x \cap V_x = \phi.$$

And, $\{\mathcal{U}_x \mid x \in A\}$ is an open subcover of A, thus it has a finite subcover $\{\mathcal{U}_{x_i} \mid i = 1, \cdots, m\}$. Finally, $\bigcup_{i=1}^{m} \mathcal{U}_{x_i}$ and $\bigcap_{i=1}^{m} V_{x_i}$ are disjoint open sets Containing A and B, respectively.

(Disjointness is given by $\left(\bigcup_{i=1}^{m} \mathcal{U}_{x_i}\right) \cap \left(\bigcap_{i=1}^{m} V_{x_i}\right) = \bigcup_{i=1}^{m} \left(\mathcal{U}_{x_i} \cap \left(\bigcap_{i=1}^{m} V_{x_i}\right)\right) = \bigcup \phi = \phi$.

「6. If $f : X \to Y$ is Continuous map where X is Compact and Y is Hausdorff, then f is closed map.

Proof. Let $C \subseteq X$ be a closed set. Since X is compact, C is Compact. And, the Continuous map f preserves Compactness, $f[C]$ is Compact in T_2, thus Closed.

⌈7. If Y is Compact, then the projection $P_X : X \times Y \to X$ is a closed map.

Proof. Let $C \subseteq X \times Y$ be a Closed set. Want to Show that $X \setminus P_X[C]$ is open.
Let $x \in X \setminus P_X[C]$ be given. Then, $(\{x\} \times Y) \cap C = \phi$, therefore $\{x\} \times Y \subseteq (X \times Y) \setminus C$.
Now, by the Tube Lemma, there is an open $\mathcal{U} \subseteq X$ s.t

$$\{x\} \times Y \subseteq \mathcal{U} \times Y \subseteq (X \times Y) \setminus C.$$

Now, $x \in \mathcal{U}$ and $\mathcal{U} \subseteq X \setminus P_X[C]$, thus $X \setminus P_X[C]$ is open.
Note: the Converse is true, but so hard to prove.

⌈Recall: The Tube Lemma.
Let X be a Topological Space, and Y be a Compact Space.
Then, for the product space $X \times Y$, for fixed $x_0 \in X$,

For any open $N \subseteq X \times Y$ s.t $\{x_0\} \times Y \subseteq N$, there is open $W \in \mathcal{T}_X$ s.t $\{x_0\} \times Y \subseteq W \times Y \subseteq N$.

Proof. Since $\{x_0\} \times Y \to Y : (x_0, y) \mapsto y$ is Homeo, $\{x_0\} \times Y$ is compact.
Since N is open, for any $y \in Y$, $\exists\, \mathcal{U}_y \in \mathcal{T}_X$, $V_y \in \mathcal{T}_Y$ s.t $\{x_0\} \times Y \subseteq \mathcal{U}_y \times V_y \subseteq N$.
Now, open cover $\{\mathcal{U}_y \times V_y \mid y \in Y\}$ of $\{x_0\} \times Y$ has a finite subcover

$$\{\mathcal{U}_{y_i} \times V_{y_i} \mid i = 1, \cdots, N\}.$$

Set $W = \bigcap_{i=1}^{N} \mathcal{U}_{y_i}$. Then, $\{x_0\} \times Y \subseteq W \times Y$ is clear, and

$$(x, y) \in W \times Y \iff \forall i = 1, \cdots, N,\ x \in \mathcal{U}_{y_i} \text{ and } \exists j\ y \in V_{y_j}$$

$$\Rightarrow (x, y) \in \mathcal{U}_{y_j} \times V_{y_j} \subseteq N.$$

8. Let X be any space, Y be a compact Hausdorff. Then,

$$f: X \to Y \text{ is continuous } \underset{\xleftarrow{\text{compact}}}{\overset{\xrightarrow{T_2}}{\text{if and only if}}} G_f = \{(x, f(x)) \mid x \in X\} \subseteq X \times Y \text{ is closed.}$$

proof. Suppose that f is continuous. Let $(x, y) \in (X \times Y) \setminus G_f$ be given. Clearly $y \neq f(x)$. Since Y is T_2, there exist disjoint opens $\mathcal{U}, V \subseteq Y$ containing $y, f(x)$, respectively. In other words,

$$(y, f(x)) \in \mathcal{U} \times V \subseteq (Y \times Y) \setminus \Delta(Y)$$

And, $f(x) \in V$ implies $x \in f^{-1}[f[\{x\}]] \subseteq f^{-1}[V] \subseteq X$. Now enough to show that only (※).

$$(x, y) \in f^{-1}[V] \times \mathcal{U} \overset{(※)}{\subseteq} (X \times Y) \setminus G_f$$

because f is continuous, thus $f^{-1}[V]$ is open. The inclusion (※) equals to:

$$\left(f^{-1}[V] \times \mathcal{U}\right) \cap G_f = \emptyset$$

and it is true. if it is not empty set, then for some $(x, f(x)) \in G_f$, $(x, f(x)) \in f^{-1}[V] \times \mathcal{U}$. That is, $f(x) \in V$ and $f(x) \in \mathcal{U}$, which is contradiction.
(Note: in this direction, we used only T_2 condition.)
Conversely, suppose that the graph G_f is closed set.
Let $\mathcal{U} \subseteq Y$ be an open set. Want to show that: $f^{-1}[\mathcal{U}]$ is an open set of X.
Let $x \in f^{-1}[\mathcal{U}]$ be given. Then, $f(x) \in \mathcal{U}$ by definition.
Let $x_0 \in X$ be given. Then, for any open $V \subseteq Y$ containing $f(x_0)$,

$$G_f \cap \left(X \times (Y \setminus V)\right)$$

is closed set. Now, since Y is compact, the projection $\pi_X: X \times Y \to X$ is closed map, thus

$$\pi_X\left[G_f \cap \left(X \times (Y \setminus V)\right)\right] \subseteq X \text{ is closed in } X.$$

Now, $\mathcal{U} = X \setminus \pi_X[G_f \cap (X \times (Y \setminus V))]$ is open, moreover, $x_0 \in \mathcal{U}$ because

$$x_0 \in \mathcal{U} \iff x_0 \notin \pi_X[G_f \cap (X \times (Y \setminus V))] \iff \forall (x, y) \in G_f \cap (X \times (Y \setminus V)),\ \pi_X(x, y) = x \neq x_0$$

But, since $(x_0, f(x_0)) \notin (X \times (Y \setminus V))$, the last asserion holds. Now, only showing $f[\mathcal{U}] \subseteq V$ is remained.
Let $x \in \mathcal{U}$. Then, $f(x) \notin Y \setminus V$, that is, $f(x) \in V$. ∎

T9. The Generalized Tube Lemma.

Let $A \subseteq X$, $B \subseteq Y$ be compact subspaces. Then,

For any open $N \subseteq X \times Y$ containing $A \times B$, there exists opens $\mathcal{U} \subseteq X$ and $V \subseteq Y$ s.t. $A \times B \subseteq \mathcal{U} \times V \subseteq N$

Proof.

T12. Let $p : X \to Y$ be a closed continuous onto map s.t. for all $y \in Y$, $p^{-1}[\{y\}]$ is compact.

If Y is compact, then X is compact. (Such p is called perfect map.)

Proof. Technically, we can write

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{y \in Y} \{y\}\right] = \bigcup_{y \in Y} p^{-1}[\{y\}]$$

Let $\mathcal{O} = \{\mathcal{U}_\alpha \mid \alpha \in \Lambda\}$ be an open cover of X. For each $y \in Y$,

$$p^{-1}[\{y\}] \subseteq \bigcup_{y \in Y} p^{-1}[\{y\}] \subseteq \bigcup_{\alpha \in \Lambda} \mathcal{U}_\alpha$$

And by the assumption, $p^{-1}[\{y\}]$ is compact, therefore there is a finite index $\Lambda_y \subseteq \Lambda$ s.t.

$$p^{-1}[\{y\}] \subseteq \bigcup_{i \in \Lambda_y} \mathcal{U}_i .$$

Now, since p is closed map,

$$V_y = Y \setminus p\left[X \setminus \left(\bigcup_{i \in \Lambda_y} \mathcal{U}_i\right)\right]$$

is an open set of Y, containing y. Now, since Y is compact, for some $y_1, \cdots, y_n$,

$$Y = \bigcup_{y \in Y} V_y = \bigcup_{j=1}^{n} V_{y_j} \Rightarrow X = p^{-1}[Y] = p^{-1}\left[\bigcup_{j=1}^{n} V_{y_j}\right] = \bigcup_{j=1}^{n} p^{-1}[V_{y_j}]$$

Meanwhile, $p^{-1}[V_y] = p^{-1}\left[Y \setminus p\left[X \setminus \bigcup_{i \in \Lambda_y} \mathcal{U}_i\right]\right] = X \setminus p^{-1}\left[p\left[X \setminus \bigcup_{i \in \Lambda_y} \mathcal{U}_i\right]\right] \subseteq \bigcup_{i \in \Lambda_y} \mathcal{U}_i$,

therefore $Y \subseteq \bigcup_{j=1}^{n} \bigcup_{i \in \Lambda_{y_j}} \mathcal{U}_i$

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Γ12. Let $p: X \to Y$ be a closed continuous onto map s.t. for all $y \in Y$, $p^{-1}[\{y\}]$ is Compact.
If Y is compact, then X is Compact. (Such p is called perfect map.)
Proof. Let $\mathcal{O} \subseteq \mathcal{T}_X$ be an open cover of X. That is, $\bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} = X$
Now, by the surjectivity, $p\left[\bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U}\right] = \bigcup_{\mathcal{U} \in \mathcal{O}} p[\mathcal{U}] = Y$.

And, since p is closed onto map, it is open map. Therefore, $p[\mathcal{U}]$ $(\mathcal{U} \in \mathcal{O})$ is open cover of Y, thus it has finite subcover s.t. $\bigcup_{i=1}^{N} p[\mathcal{U}_i] = Y$, since Y is compact. Observe that

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{i=1}^{N} p[\mathcal{U}_i]\right] = \bigcup_{i=1}^{N} p^{-1}[p[\mathcal{U}_i]]$$

Let $y \in Y$ be given. By the surjectivity, $p^{-1}[\{y\}]$ is non-empty.
Let $x_0 \in X$ s.t. $p(x_0) = y$.

$$p^{-1}[W] \subseteq \mathcal{U}$$
$$\Leftrightarrow W \subseteq p[\mathcal{U}]$$

Let $y \in Y$ be given. If $\mathcal{U} \subseteq X$ is open containing $p^{-1}[\{y\}]$, then

$$p^{-1}[\{y\}] \subseteq \mathcal{U} \Rightarrow p[p^{-1}[\{y\}]] \subseteq p[\mathcal{U}] \Rightarrow y \in p[\mathcal{U}]$$
$$\Rightarrow y \in W \subseteq p[\mathcal{U}]$$

$$Y = \bigcup_{y \in Y} \left(Y \setminus p\left[X \setminus \left(\bigcup_{i \in \Lambda_y} \mathcal{U}_i\right)\right]\right) = \bigcup_{j=1}^{n} \left(Y \setminus p\left[X \setminus \left(\bigcup_{i \in \Lambda_{y_j}} \mathcal{U}_i\right)\right]\right)$$